MOHIT KHAIRNAR

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TECHNICAL SKILLS

Generative AI & LLMs: LLMs, Agentic AI Frameworks, RAG Architectures, Prompt Engineering, LangChain, LlamaIndex, OpenAI API, Hugging Face Transformers

AI/ML & Data: Python, TensorFlow, Scikit-learn, Spark, SQL, PL/SQL, AI/ML Pipelines, Data Preparation & ETL Workflows

Cloud & Platforms: AWS Certified, Databricks, Docker, Kubernetes, CI/CD

MLOps / LLMOps & Responsible AI: Model Deployment & Monitoring, AI Governance, Bias Detection Awareness, Hallucination Mitigation, Output Safety Guardrails, Harness, Explainability Concepts

Vector Databases & Integration: Pinecone, Qdrant, FAISS, REST APIs, Microservices, Apache Kafka (Event Streaming)

Programming & Frameworks: Python, JavaScript, Java, C++, C# (.NET), Django, FastAPI, ExpressJS, ReactJS

Databases & Version Control: MySQL, Oracle PL/SQL, MongoDB, Git

PROFESSIONAL EXPERIENCE

AI & Backend Engineer (Backend Developer) *Aug 2024 - Present*
Tata Consultancy Services (TCS) Mumbai, Maharashtra

- Designed and implemented Generative AI and Retrieval-Augmented Generation (RAG) solutions using LangChain, LlamaIndex, and the OpenAI API, integrating vector databases (Pinecone) to power semantic search and automate structured/unstructured data ETL pipelines, improving resolution time by 40%.
- Built and maintained scalable AI/ML data pipelines and cloud-native integrations on AWS, supporting ingestion and transformation of 50+ MB of trade data per day across 2M+ daily stock transactions on NSE/BSE using optimized PL/SQL procedures.
- Integrated AI capabilities with internal APIs, databases, and enterprise applications, engineering real-time event-driven data pipelines with Apache Kafka to enable low-latency, scalable streaming data processing.
- Supported deployment, monitoring, and operational maintenance of production AI and backend services using Docker and AWS, following CI/CD and reliability best practices aligned with MLOps principles.
- Developed high-performance microservices in .NET (C#) and Python using concurrency and asynchronous processing patterns, improving API performance, data throughput, and batch processing time by 40%.
- Maintained 99% system availability across 6+ banking brokers by managing zero-downtime production releases, enforcing version control (Git) discipline, and resolving critical issues within 1-hour SLAs.
- Built evaluation pipelines/harnesses to benchmark LLM outputs against golden datasets, tracking metrics like accuracy, relevance, and hallucination rate.
- Fine-tuned/instruction-tuned open-source LLMs (LLaMA, Mistral) using LoRA/QLoRA for domain-specific tasks, reducing inference cost for calling larger foundation models.

RELEVANT PROJECTS

Enterprise RAG Knowledge Assistant — *Python, LangChain, OpenAI API, FAISS, FastAPI*

- Built an end-to-end Retrieval-Augmented Generation assistant that grounds LLM responses in internal documents via a FAISS vector store, reducing hallucinated answers and improving response accuracy for domain-specific queries.
- Implemented prompt engineering and output validation guardrails to improve response safety and reliability, and exposed the assistant via a FastAPI service for integration with downstream applications.

Workotopia - AI Recruiter — *Python, LangChain, Streamlit, Hugging Face*

- Built an AI-powered recruiting agent using BERT embeddings and cosine similarity to score resumes against job descriptions, structuring an early agentic workflow for multi-step candidate evaluation.

Magna - AI Agriculture Platform — *Python, Scikit-learn, Flask, Firebase, React*

- Developed an IoT-enabled crop and soil monitoring platform with a cloud-integrated data pipeline and React dashboard, training ensemble ML models that achieved 95% prediction accuracy.

EDUCATION

Bachelor of Engineering in Information Technology *Jun 2021 - Mar 2024*
Sardar Vallabhbhai Patel Institute of Technology, Anand Gujarat GPA: 8.5/10

Diploma in Information Technology *Apr 2018 - Mar 2021*
Parul Polytechnic Institute of Technology, Vadodara Gujarat GPA: 9.6/10

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner